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Dept. of Computer Information Systems

CIS2091 -- PDBMS Practical -- Temporal

Instructions: Work in groups but no inter group sharing of ideas, solutions, or hints. If you use a manual (or any resource) not mentioned in assignment, you need to ref it. Questions marked with an asterisk () are *OPTIONAL* -- i.e., no marks deducted if not attempted.*

You are allowed to use 'range of' data type. In which case, it is important to describe the DDL and DML required.

Start-up a postgresql server instance. Also make sure you have the Scott database available. (The following was based on versions 9.5 and later – execute `SELECT version();` on an SQL shell to determine your version).

Study the following sections from PostgreSQL manual (online): Date/Time Types, Date/Time Intervals, Date/Time Functions and Operators. Please also revise the functions `case`, `overlaps`, `to_char`, `to_number`, `to_date`, `interval`, and `extract`.

Note that the table `emp` in schema `scott` has a date typed attribute – called `hiredate`.

Q1 Calculate the span of recruitment dates (note in month & year)? Output should be similar to the following.

```
"from"; "Dec-1980"; "to"; "Jan-1983"
```

Q2 Which employee has been employed first and who was the last?

Q3 The difference in days in recruitment when is compared to that of the 'president'. Sort output.

Q4 For those employees recruited before the 'president', compute the number of months in employment prior to his engagement. The following output is meant to guide you.

```
Days,years,months,comp months;
387;1;1;13;
421;2;-10;14;
16;0;1;1;
16;0;1;1;
67;1;-10;2;
```

Q5 Compute the number of **working days** for each employee from their recruitment date to, say 13th December 1985. *State any assumptions and applicability of your computation.*

Q6 Create a schema, e.g. temporal, and a table with the following relational structure:

EMPLOYEE(Serial No., Name, Salary, Title, BirthDate, StartDate, EndDate).

Populate the relation with some data (at least as found in table one below).

Q7 Find the “current” salary for Charlie from employee. (*Assume that current date is 13th December 1995*).

Q8 We want to access the salary history of an employee (e.g. Charlie) and produce result similar to table two (coalesce tuples that are continues in time into a single one). Write a procedure, in your favourite programming language, as a first cut solution to this query – example output for employee Charlie in found in table two below.

Q9* Write down the salary history query of an employee (e.g. Charlie) as a stored procedure using cursors.

Q10* Could we write an SQL construct (i.e. using only SQL) for the same salary history query? If so write it down (or sketch it). Incomplete or non-general solution is acceptable.

Q11 Consider the following approach to solve the salary history query. Let us split the second relation schema into two relations:

EMPLOYEE1(Serial No.,Name, Salary, BirthDate, StartDate, EndDate); and

EMPLOYEE2(Serial No.,Name, Title, BirthDate, StartDate, EndDate).

What are the basic consequences of this approach?

How would you write the salary and department history for any employee?

Table One: An Employee’s View of Salary and Job title history

	<i>Name</i>	<i>Salary</i>	<i>Title</i>	<i>BirthDate</i>	<i>StartDate</i>	<i>StopDate</i>
103	CHARLIE	5000	Teacher	1960-Jan-05	1995-Jan-01	1995-Jun-01
235	CHARLIE	6000	Teacher	1960-Jan-05	1995-Jun-01	1995-Oct-01
765	CHARLIE	6000	Form Master	1960-Jan-05	1995-Oct-01	1996-Feb-01
769	CHARLIE	6000	Asst Head	1960-Jan-05	1996-Feb-01	1997-Jan-01

Table Two: Charlie’s Salary History (Query Result)

<i>Name</i>	<i>Salary</i>	<i>Start</i>	<i>Stop</i>
CHARLIE	5000	1995-Jan-01	1995-Jun-01
CHARLIE	6000	1995-Jun-01	1997-Jan-01

